

DYQ-7 智能液位计 使用说明书 V1.0 Instructions for DYQ-7 Intelligent Level Gauge

成都兰石低温科技有限公司 2020-7-9

Chendu Lanshi Cryogenic Technology Co., Ltd. Jul. 9, 2020

1、概述

1. Overview

DYQ-7 是成都兰石低温科技有限公司第七代智能液位计，采用全新的模块化方案设计，每个传感器采集单元都相互独立且互不影响，减少了测量误差，提高了产品测量精度与产品可靠性，同时解决了上一代因产品型号众多，配置复杂等带来的生产、订货管理等一系列问题。DYQ-7 is the seventh generation intelligent level gauge of Chengdu Lanshi Cryogenic Technology Co., Ltd. With a new modular design, sensor acquisition units are independent to each other and do not affect each other, reducing measurement errors, and improving product measurement accuracy and product reliability. In addition, it solves a series of problems in production and order management of the previous generation due to numerous product models and complex configurations.

DYQ-7 系列智能液位计由变送器、电容测量单元、压力采集单元、温度测量单元、燃气报警监控单元、真空监测单元、DTU 单元等组成，客户只用按需选择对应功能模块，我司即可快速交付相关产品。

DYQ-7 Intelligent Level Gauge is composed of a transmitter, a capacitance measurement unit, a pressure acquisition unit, a temperature measurement unit, a gas alarm monitoring unit, a vacuum monitoring unit, a DTU, etc. Customers only need to select the corresponding functional module according to their needs, and we can quickly deliver related products.

2、技术参数

2. Technical Parameters

组成部件 Component	项目 Item	型号 Model			备注 Remarks
		DYQ-7C	DYQ-7A	DYQ-7D	
变送器 Transmitter	供电电压 Power supply voltage	15~36VDC	15~36VDC	15~36VDC	
	电流 (Imax) Current (Imax)	400mA	400mA	400mA	
	工作温度 Operating temperature	-40~85℃	-40~85℃	-40~85℃	
	外形尺寸	109*92*49mm	109*92*49mm	109*92*49mm	

	Dimensions				
	重量 Weight	400g	460g	400g	
	通信接口 1 Communication interface 1	5 芯利路通母端 LLT 5-core female	5 芯利路通母端 LLT 5-core female	5 芯利路通母端 LLT 5-core female	
	通信接口 2 Communication interface 2	AMP 282088-1	AMP 282087-1	AMP 282088-1	选配 Optional
		AMP 282106-1			
	通信方式 Communication mode	CAN	0~4.2V	RS485	
	防护等级 Protection grade	IP67	IP67	IP67	
	防爆等级 Explosion-proof grade	Ex icⅡA T4 Gc	Ex icⅡA T4 Gc	Ex icⅡA T4 Gc	
	电磁兼容 EMC	ECE R10	ECE R10	ECE R10	
	型式试验依据 Type test basis	GB/T 28046	GB/T 28046	GB/T 28046	
I 电缆 I-type cable	接口形式 Interface type	5 芯利路通母端 LLT 5-core female	5 芯利路通母端 LLT 5-core female	5 芯利路通母端 LLT 5-core female	
	长度 Length	600mm	600mm	600mm	
	L 型连接器 L-type connector	整体集成 Overall integration	整体集成 Overall integration	整体集成 Overall integration	
	测量精度 Measuring accuracy	±3%FS	±3%FS	±3%FS	
压力传感器 Pressure sensor	接口形式（输入） Interface type (input)	1/4 NPT 外螺纹 1/4 NPT male thread	-	1/4 NPT 外螺纹 1/4 NPT male thread	
	接口形式（输出） Interface type (output)	5 芯利路通母端 LLT 5-core female	-	5 芯利路通母端 LLT 5-core female	
	传感器类型 Sensor type	电流型 Current type	-	电流型 Current type	

	测量范围 Measuring range	0~4 MPa	-	0~4 Mpa	选配 Optional
	测量精度 Measuring accuracy	±0.5%FS	-	±0.5%FS	
温度传感器 Temperature sensor	接口形式（输入） Interface type (input)	1/4 NPT 外螺纹 1/4 NPT male thread	-	1/4 NPT 外螺纹 1/4 NPT male thread	
	接口形式（输出） Interface type (output)	5 芯利路通母端 LLT 5-core female	-	5 芯利路通母端 LLT 5-core female	
	传感器类型 Sensor type	PT1000	-	PT1000	
	测量范围 Measuring range	-196~150℃	-	-196~150℃	
	测量精度 Measuring accuracy	±1℃	-	±1℃	
燃气报警器 Gas alarm	接口形式（输出） Interface type (output)	5 芯利路通母端 LLT 5-core female	-	5 芯利路通母端 LLT 5-core female	
	安装方式 Mounting way	专用安装版 Special Mounting Version	-	专用安装版 Special Mounting Version	
	报警浓度 Alarm concentration	7000ppm	-	7000ppm	
	允许误差 Permissible error	<3% LEL	-	<3% LEL	
	响应时间 Response time	≤10S	-	≤10S	
	防护等级 Protection grade	IP65	-	IP65	
真空规管 Vacuum gauge	接口形式（输出） Interface type (output)	5 芯利路通母端 LLT 5-core female	-	5 芯利路通母端 LLT 5-core female	
	安装方式 Mounting way	自密封 Self-sealing	-	自密封 Self-sealing	
	测量范围	0.01~100Pa	-	0.01~100Pa	

DTU	Measuring range 测量精度 Measuring accuracy	±20%	-	±20%
	传输方式 Transmission mode	4G	-	4G
	传输频率（可调） Transmission frequency (adjustable)	5 分钟/次 5min/time	-	5 分钟/次 5min/time
	定位方式 Positioning mode	GPS/北斗 GPS / Beidou	-	GPS/北斗 GPS / Beidou
	定位精度 Positioning accuracy	<5 米 <5m	-	<5 米 <5m
说明：电容式传感器根据客户气瓶尺寸选购； Note: The capacitive sensor is selected according to the size of the customer's cylinder.				

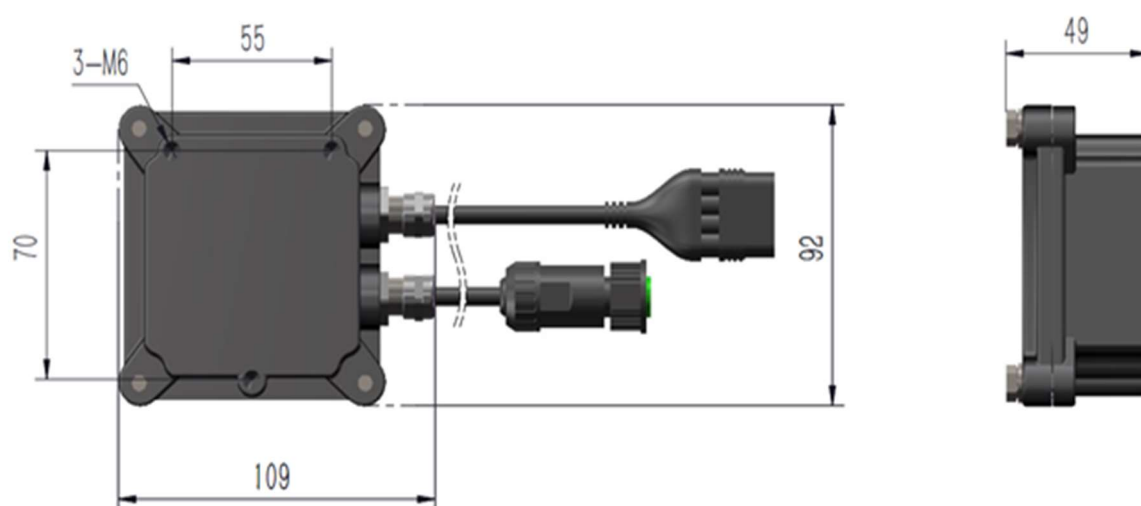
3、配件清单

3. List of Accessories

序号 No.	名称 Description	数量 Qty.	备注 Remarks
1	变送器 Transmitter	1	
2	I 电缆 I-type cable	1	
3	安装板 Mounting plate	1	
4	安装螺钉 Mounting screw	4	M6
5	配套线缆 Supporting cable	1	
6	使用说明书 User's Manual	1	
7	检验报告 Inspection Report	1	
8	合格证 Certificate of Conformity	1	

4、产品尺寸

4. Product Dimensions



5、显示说明

Display Description

DYQ-7C 显示器可支持多至 4 个气瓶的数据采集和显示。用户可以根据需要选用 94mm 圆形显示器（图示液位、缓冲罐压力与气瓶压力值仅作参考）。

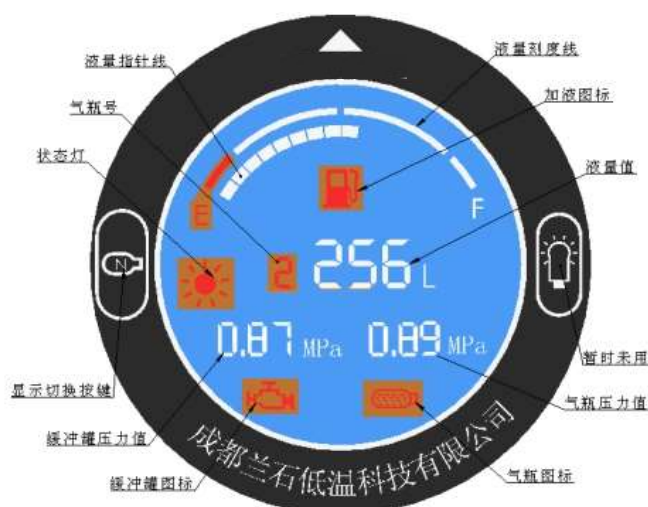
DYQ-7C display can support data collection and display of up to 4 cylinders. The user can choose a 94mm circular display (the liquid level, buffer tank pressure and cylinder pressure shown in the figure are for reference only).

显示分为 5 个部分

- ▼ 液量刻度显示部分
- ▼ 液量数字显示部分
- ▼ 气瓶压力数字显示部分
- ▼ 缓冲罐压力数字显示部分
- ▼ 气瓶号及状态灯显示部分

There are five display parts.

- ▼ Display of liquid volume scale
- ▼ Digital display of liquid volume
- ▼ Digital display of cylinder pressure
- ▼ Digital display of buffer tank
- ▼ Display of cylinder number and status light



5.1. 液量刻度显示部分:

▼ 液量刻度显示部分由刻度线和指针组成。刻度线是气瓶有效容积的线性分度。指针指示气瓶内可用 LNG 的相对容量值。

▼ 当指针达到刻度线的 F 时, 表示气瓶内液位达到满刻度。

当指针达到刻度线的 E 时, 表示气瓶内液位降到零刻度。

当气瓶内剩余 LNG 容量下降到红色刻度区域以内时, 表示气瓶内剩余 LNG 已经很少, 提示用户及时加液 (圆形显示器状态灯将闪烁告警)。

Display of liquid volume scale:

▼ This part is composed of scale line and indicator. The scale line is a linear division of the effective volume of the cylinder. The pointer indicates the relative capacity value of the available LNG in the cylinder.

▼ When the pointer reaches F on the scale line, it indicates that the liquid level in the cylinder has reached full scale.

When the pointer reaches E on the scale line, it indicates that the liquid level in the cylinder has dropped to zero.

When the remaining LNG capacity in the cylinder drops to within the red scale area, it indicates that there is little remaining LNG in the cylinder and prompts the user for filling in time (the status light of the circular display will flash for alarm).

5.2. 液量数字显示部分

▼ 液量显示由 3 位数字组成, 显示气瓶内可用的 LNG 容量, 单位为升 (L)。

▼ 液量显示

正常状态: 显示数字表示气瓶内可用的 LNG 容量, 单位为升。

满液状态: 当显示值为 "FULL" 时, 表示气瓶内液量达到满刻度。

下溢出状态: 当液量值显示 "- EE" 时, 表示下溢出, 多数情况是液位传感器未连接上造成的错误。

上溢出状态: 当液量值显示 "+ EE" 时, 表示上溢出, 多数情况是液位传感器短路或插接器进水造成的错误。

注 1: 这里的零刻度和满刻度分别是指液位传感器检测范围内的最小值和最大值。

注 2: 液位达到零刻度时, 气瓶内剩余 LNG 大约为气瓶标称容积的 6%, 但不能精确保证。达到加液告警时, 用户应尽快加液。

Digital display of liquid volume

▼ The display of liquid volume is composed of 3 digits, which displays the available LNG capacity in the cylinder, in liter (L).

▼ Display of liquid volume

Normal status: The digits indicate the available LNG capacity in the cylinder, in liter.

Full liquid status: When the displayed value is "FULL", it indicates the liquid volume in the cylinder has reached the full scale.

Underflow status: When the liquid volume value is "- EE", it indicates underflow, which is mostly an error caused by the liquid level sensor not being connected.

Overflow status: When the liquid volume value is "+ EE", it indicates overflow, which is mostly an error caused by short circuit of the liquid level sensor or water in the connector.

Note 1: The zero scale and full scale here respectively refer to the minimum and maximum values within the detection range of the liquid level sensor.

Note 2: When the liquid level reaches zero scale, the remaining LNG in the cylinder is about 6% of the nominal volume of the cylinder, but it cannot be accurately guaranteed. When the liquid filling alarm is made, the user

shall add liquid as soon as possible.

5.3. 压力显示部分

▼ 气瓶压力数字显示： 3 位数字，小数点后 2 位，单位 MPa，最大量程 4MPa。

▼ 缓冲罐压力数字显示： 3 位数字，小数点后 2 位，单位 MPa，最大量程 4MPa。

▼ 压力显示

正常状态： 数字表示压力值，单位为 MPa。

下溢出状态： 显示值为“- EE”， 多数情况是由于压力传感器损坏造成。

上溢出状态： 显示值为“+ EE”， 多数情况是由于压力传感器损坏造成。

未接入状态：显示值为“—”，表示用户未接入压力传感器。

Display of pressures

▼ Digital display of cylinder pressure: 3 digits, 2 digits after the decimal point, in MPa, maximum range 4 MPa.

▼ Digital display of buffer tank pressure: 3 digits, 2 digits after the decimal point, in MPa, maximum range 4MPa.

▼ Pressure display

Normal status: The digits indicate the pressure value, in MPa.

Underflow status: The displayed value is "- EE", which is mostly caused by damage to the pressure sensor.

Overflow status: The displayed value is "+ EE", which is mostly caused by damage to the pressure sensor.

No-connection status: The displayed value is "---", which indicates that the user is not connected to the pressure sensor.

5.4. 气瓶号及状态灯显示部分

▼ 气瓶号显示

显示值为 1~4，对应 1~4 号气瓶编号，表示当前屏上的显示值所对应的气瓶编号。用户可以通过圆形显示器左边的切换按键或触摸矩形显示屏任何位置选择要显示的气瓶。

▼ 状态灯显示（圆形显示器）

闪烁： 当气瓶内剩余 LNG 高度下降到加液告警门限以下时，状态灯闪烁告警，提示用户及时加液。告警门限大约为气瓶标称容积的 6%

常亮： 未正确连接变送器。

Display of cylinder number and status light

▼ Display of cylinder number

The displayed value is 1~4, corresponding to the cylinder number 1~4, indicating the cylinder number corresponding to the value displayed on the current screen. The user can select the cylinder to be displayed through the switch button on the left side of the circular display or touch any position of the rectangular display.

▼ Display of status light (circular display)

Flashing: When the height of the remaining LNG in the cylinder drops below the filling alarm threshold, the status light flashes to alert the user to fill up in time. The alarm threshold is approximately 6% of the nominal volume of the cylinder.

Constant ON: The transmitter is not connected properly.

6、液位计安装

Mounting of Level Gauge

6.1、使用工具清单

List of Tools

序号 No.	名称 Description	规格型号 Spec. & Model	数量 Qty.	备注 Remarks
1	扳手 Wrench	6#	1	
2	绑扎带 Banding tape		1	
3	斜口钳 Diagonal pliers		1	

6.2、安装步骤

Mounting Steps

用 M6 螺钉将 L 型支架（宽边）与变送器紧固，然后将装好支架的变送器支架另一端（窄边）与车辆上变送器安装孔位进行对齐并用 M6 螺钉紧固即可。

Fasten the L-shaped support (wide side) with the transmitter with M6 screws, then align the other end (narrow side) of the mounted transmitter support with the transmitter mounting hole on the vehicle and fasten them with M6 screws.

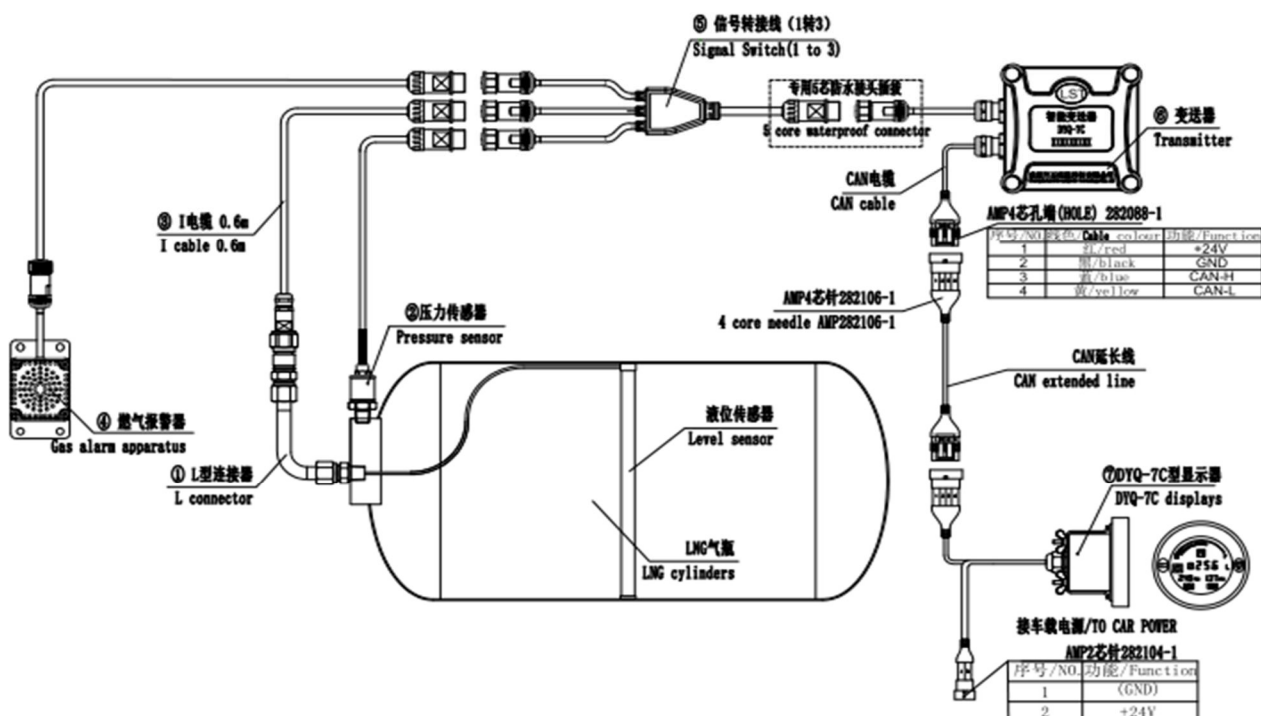


6.3、应用案例

Application cases

以具备燃气报警、压力测量功能的智能液位计为例，接线示意如下图所示，现场实际应用连接根据客户具体的选型确定，增减对应连线即可。

Take an intelligent level gauge with gas alarm, and pressure measurement functions as an example. The wiring is shown in the following figure. The actual wiring on site should be determined according to the customer's specific type selection, and the corresponding wiring can be increased or decreased.



7、操作与维护

Operation and Maintenance

7.1、上电自检

Power-on self-inspection

颜色 Color		状态 Status	说明 Description
红灯 Red light	常亮 Normally on	上电初始态/传感器未接入 Power-on initial state/sensor not connected	

	闪烁 Flashing	传感器故障 Sensor fault	
蓝灯 Blue light	常亮 Normally on	配置器接入 Configurator connected	
	闪烁 Flashing	变送器正在升级 Transmitter being upgraded	
绿灯 Green light	常亮 Normally on	工作正常 Normal working	

7.2、I 电缆配置

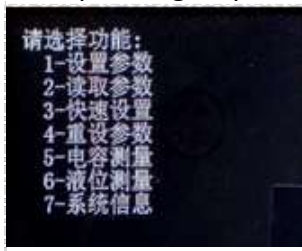
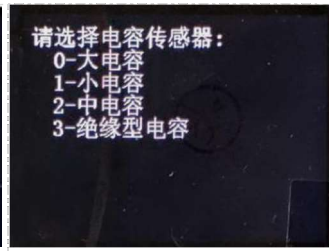
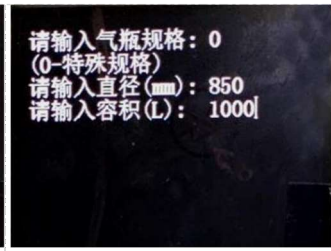
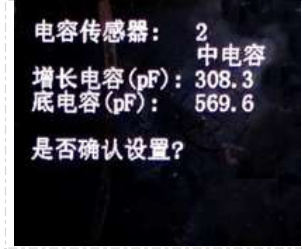
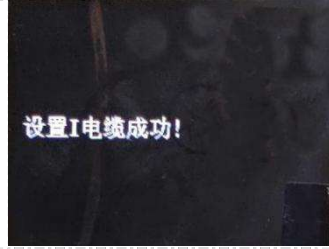
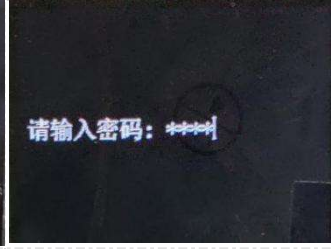
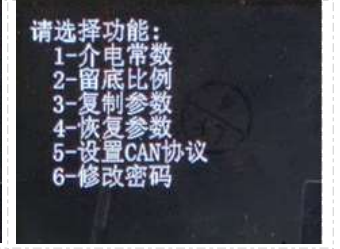
Configuration of I-type cable

- ① 将 I 电缆与配置器相连；
- ① Connect the T-type cable with the configurator.
- ② 根据所接气瓶规格类型，进行底电容、介电常数等参数的配置下发；
- ② **Configure and issue parameters such as bottom capacitance and dielectric constant according to the specifications and types of the connected cylinder.**
- ③ 下述以直径 850mm，容积 1000L 气瓶为例，进行模拟配置说明；
- ③ Take a cylinder with a diameter of 850mm and a volume of 1,000L as an example to describe the simulated configuration:



操作步骤，如下图描述所示：

The operating steps are shown as follows:

			
☑ 按『开/关机键』进入功能界面，键入『1』选中『设置参数』 ☑ Press "On/Off" to enter the function interface, and enter "1" to select "Set Parameters"	☑ 键入『2』选中『中电容』 ☑ Enter "2" to select "Neutralizing Capacitor"	☑ 依次键入并按『确认键』输入气瓶规格、直径、容积 ☑ Enter the information of cylinder, diameter and volume one by one and press "OK"	☑ 确认输入完成后，进入配置参数预览界面第 1 页 ☑ Enter Page 1 of the configuration parameter preview interface
			
☑ 按『确认键』进入参数预览第 2 页 ☑ Press "OK" to enter Page 2 of the parameter preview interface	☑ 按『确认键』完成 I 电缆配置 ☑ Press "OK" to complete the I-type cable configuration	☑ 在开机界面按『设置键』，键入默认密码『2222』 ☑ Press "Setting" on the start interface and enter the default password "2222"	☑ 确认进入更多参数的配置页面。如非必要，此菜单慎改 ☑ Press "OK" and enter the configuration interface of more parameters. Do not change this menu if not necessary.

8、验收与存放

Acceptance and Storage

- ① 用户根据配件清单对产品及其附件进行验收；
- ① The customer should check and accept the product and its accessories according to the list of accessories.
- ② 打开包装检查变送器等外观是否完好无损，如有问题请及时联系我司；
- ② The customer should open the package to check whether the appearance of the transmitter is intact. Please contact us in time in case of any problem.
- ③ 原包装按照产品规格书要求进行存放，相对湿度<80%的室内环境；
- ③ The original packaging shall be stored in accordance with the requirements of the product specification in an indoor environment with a relative humidity of <80%.

9、售后服务

After-sales Services

成都兰石低温科技有限公司热忱服务用户，欢迎用户反馈产品使用过程中的相关信息，用户在安装、使用、效验中遇有疑问，公司将及时给予技术支持。

We, Chengdu Lanshi Cryogenic Technology Co., Ltd., are dedicated to serving our customers and invite customers to feed back relevant information during the use of our products. We will provide timely technical support if our customers encounter any questions during mounting, use, and validation.

服务热线： 028-85739020

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